

On the Limits of Temporal Sampling in Video Action Recognition: Strategies for Sensing and Modeling in Human Activity Perception

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Abstract. Temporal resolution in video action recognition is often treated as a fixed hyperparameter rather than a signal processing constraint. We present a systematic analysis of temporal aliasing across benchmark datasets (UCF101, Kinetics-400) and transformer architectures (TimeS-former, VideoMAE, ViViT) using a controlled coverage-by-stride sampling protocol. Our results reveal that accuracy degrades monotonically with reduced temporal evidence, but the severity is governed by class-level motion frequency and architectural priors. High-frequency motion events collapse under sparse sampling, while slower or context-dominated activities remain more robust. Notably, VideoMAE exhibits extreme sensitivity with a 17.2% drop on UCF101 at 25% coverage, whereas TimeS-former’s factorized attention proves more resilient (6.9% drop). We show that undersampling induces significant variance inflation, signaling instability beyond mean accuracy. To mitigate this, we introduce Temporal Robustness Augmentation (TRA), a training strategy using randomized temporal configurations, which improves robustness in sparse regimes (up to ~5% relative improvement) without compromising dense-sampling performance. Finally, we provide an analysis tool with tabularized findings and practical insights for sensor selection and model training, available in anonymized form for peer review. Our findings demonstrate that while temporal sampling follows classical signal processing principles, aliasing effects can be mitigated through representation learning, motivating content-adaptive designs for efficient video systems.

Keywords: Temporal Aliasing, Video Action Recognition, Sampling Robustness, Video Transformers

1 Introduction

Human activity and state understanding is a foundational problem in video understanding with broad impact on autonomous systems, healthcare monitoring, and human-computer interaction. Beyond identifying explicit physical actions, temporal dynamics are critical for inferring subtle internal states, such as emotion, health status, alertness, or intoxication [2, 9]. Early progress in the field was driven by convolutional architectures that extended spatial feature learning into the temporal domain. Pioneering work such as C3D demonstrated that spatiotemporal features could be learned directly from short video clips using

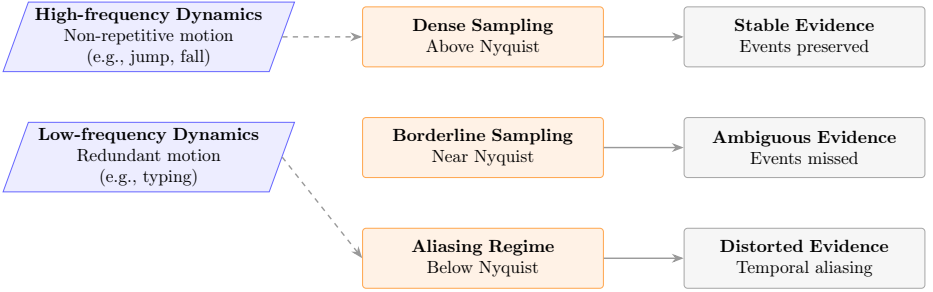


Fig. 1: Temporal sampling viewed through signal-processing principles. Human actions exhibit intrinsic temporal frequencies; when the sampling rate falls below the Nyquist limit, motion evidence becomes ambiguous or distorted despite preserved spatial content. Dashed arrows indicate sensitivity to sampling rather than causation.

three-dimensional convolutions [15]. Subsequent advances emphasized scalability and transfer learning, most notably through inflated 3D architectures (I3D) pre-trained on large-scale datasets such as Kinetics [4]. While effective, these models incurred significant computational cost and relied on fixed temporal receptive fields.

Recent years have seen a paradigm shift toward transformer-based video architectures that model long-range dependencies through attention mechanisms. TimeSformer introduced factorized space-time attention to reduce computational complexity while maintaining global temporal context [3]. ViViT explored alternative tokenization and attention factorization strategies for video transformers [1], and VideoMAE demonstrated that masked autoencoding can learn strong video representations under extreme temporal masking [13]. Despite impressive benchmark performance, these architectures are almost universally trained and evaluated using standardized temporal sampling recipes (e.g., 8, 16, or 32 uniformly sampled frames), implicitly treating temporal resolution as a static implementation detail rather than a fundamental research variable.

This assumption becomes increasingly problematic in real-world deployments. In embedded and resource-constrained environments, dense temporal sampling may be infeasible due to bandwidth, power, or latency constraints. Consequently, many practical systems adopt sparse temporal sampling or lightweight temporal modeling to reduce computational overhead [10]. However, temporal down-sampling is not merely an efficiency optimization: it fundamentally alters the information available to the recognition model. A principled understanding of *when* temporal subsampling becomes destructive and *how* different architectures respond to this regime shift remains largely absent from the literature.

Sampling theory offers a natural lens for addressing this gap. Figure 1 illustrates this signal-processing perspective, while Figure 2 provides a real visual example showing how temporal coverage and sampling stride jointly determine whether action dynamics are faithfully preserved or distorted by aliasing. The Nyquist-Shannon sampling theorem states that a signal must be sampled at least twice its highest frequency component to avoid aliasing [11]. Many human actions contain high-frequency temporal events (e.g., rapid posture transitions

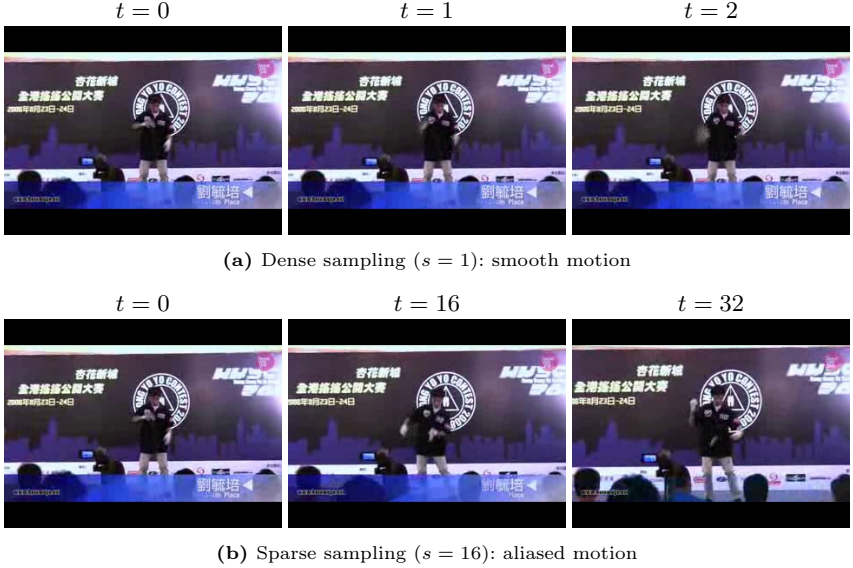


Fig. 2: Visual manifestation of temporal aliasing on the *YoYo* action class from the UCF101 dataset [12]. Sparse temporal sampling introduces discontinuities and strobing artifacts.

in a fall, the release phase of a tennis serve) whose discriminative cues may be irretrievably lost when the sampling rate drops below a critical threshold. This phenomenon, known as *temporal aliasing*, distorts motion evidence and can introduce ambiguity or perceptual reversals, exemplified by classical wagon-wheel illusions in motion perception [17]. While aliasing is well understood in signal processing and perception, its implications for modern deep video architectures remain underexplored. Related evidence from deep learning suggests that naive subsampling can violate sampling assumptions and harm robustness, as demonstrated by anti-aliasing techniques in convolutional networks [21]. Yet, in video recognition, performance is still typically reported as mean accuracy under a small set of fixed sampling choices, offering limited insight into stability and class-dependent failure modes under systematic undersampling.

In this work, we conduct a large-scale empirical study on UCF101 [12] and Kinetics-400 [8], evaluating three representative transformer paradigms: factorized attention (TimeSformer), masked reconstruction (VideoMAE), and spatiotemporal tokenization (ViViT). Our contributions are fourfold: (1) we quantify architecture-dependent brittleness to temporal aliasing, (2) we uncover strong class-dependent heterogeneity in sampling sensitivity, (3) we demonstrate that performance variance increases sharply under aggressive subsampling, revealing instability that is masked by mean accuracy alone, and (4) we introduce Temporal Robustness Augmentation (TRA), a randomized temporal training strategy that improves robustness in sparse regimes without compromising dense-sampling performance. Together, these findings provide principled guidance for optimizing spatiotemporal resolution in resource-constrained recognition sys-

tems. An implementation of the analysis tool, including tabularized findings and insights for sensor selection and model training, is available at [anonymized for peer review; see supplementary material].

2 Related Work

2.1 Deep Architectures for Action Recognition

Video action recognition has progressed from early 3D convolutional networks that jointly model space and time to transformer-based architectures with stronger long-range dependencies. Early 3D ConvNets such as C3D demonstrated that spatiotemporal features can be learned end-to-end from raw clips, but at significant computational cost [15]. Two-stream and inflated designs improved accuracy and transfer by leveraging large-scale pretraining [4]. Subsequent work investigated factorized spatiotemporal convolutions ($R(2+1)D$) to improve optimization and accuracy [16] and developed families of efficient video models that explore the accuracy–compute trade-off more systematically [5].

More recently, transformers have become dominant in video recognition. TimeSformer introduced factorized space–time attention to reduce the quadratic cost of full spatiotemporal attention while preserving global context [3]. ViViT explored alternative tokenization and attention factorization strategies for video transformers [1]. VideoMAE showed that masked autoencoding can learn strong video representations under extreme masking ratios, improving data efficiency [13]. Despite strong benchmark performance, most of these architectures are evaluated under fixed clip lengths and standardized sampling recipes (e.g., 8/16/32 frames with uniform stride), leaving their robustness to systematic changes in temporal sampling comparatively under-characterized.

2.2 Temporal Sampling, Multi-Rate Modeling, and Adaptive Inference

Temporal sampling is often treated as an implementation detail, with a fixed number of uniformly sampled frames used regardless of an action’s tempo or duration. Temporal Segment Networks (TSN) popularized sparse segment-based sampling to efficiently cover long videos, making uniform sparse sampling a common baseline [18]. Efficient temporal modeling methods such as the Temporal Shift Module (TSM) reduce computation by approximating temporal interactions through lightweight feature shifts while still relying on fixed sampling schemes [10].

Other work introduces multi-rate temporal structure. SlowFast employs dual pathways operating at different frame rates to capture semantics and motion cues [6], while Temporal Pyramid Networks (TPN) aggregate representations across multiple temporal resolutions [20]. Adaptive frame selection methods such as AdaFrame dynamically select informative frames [19]. However, these approaches primarily optimize efficiency and rarely analyze temporal downsampling through a sampling-theoretic perspective or quantify instability under aggressive undersampling.

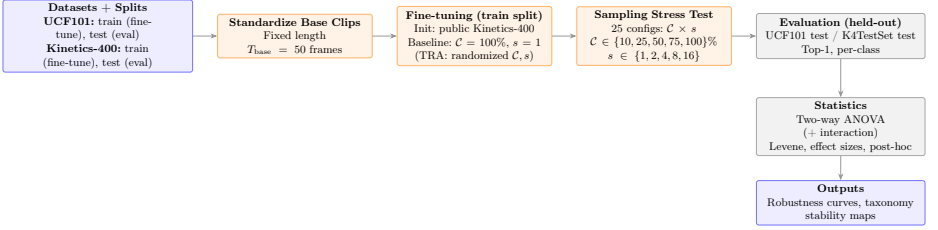


Fig. 3: End-to-end methodology. We standardize videos into fixed-length base clips, initialize from public Kinetics-400 pretrained checkpoints, fine-tune on the target dataset, and evaluate robustness over a controlled coverage $\times$ stride grid. In additional experiments, we apply TRA by randomizing coverage and stride during fine-tuning to encourage sampling-invariant representations.

2.3 Aliasing, Sampling Theory, and Frequency-Space Views of Motion

The Nyquist–Shannon theorem states that signals must be sampled above twice their highest frequency component to avoid aliasing [11]. Temporal aliasing is well documented in imaging and perception (e.g., stroboscopic motion reversal) when discrete sampling cannot resolve rapid state changes [17]. In deep learning, anti-aliasing before downsampling can improve stability and robustness in convolutional networks, suggesting that standard subsampling may violate classical sampling assumptions [21]. Frequency-domain representations have also been explored for action recognition, showing that motion and pose signals can be informative in the spectral domain [14].

Our work differs from prior efficiency- or architecture-driven studies by treating temporal downsampling as a controlled sampling process. We evaluate modern transformer paradigms under a coverage $\times$ stride sweep and quantify mean accuracy loss alongside variability and class-dependent sensitivity, linking empirical behavior to sampling-theoretic expectations.

3 Methodology

3.1 Datasets and Preprocessing

To ensure our findings generalize across different motion dynamics, we evaluate temporal sampling effects on two large-scale benchmark datasets with distinct characteristics. Figure 3 provides an end-to-end overview of the experimental pipeline, illustrating how standardized video clips are sampled under controlled coverage and stride configurations, processed by pretrained and fine-tuned transformer models, and evaluated using both accuracy and stability metrics.

UCF101 [12] serves as our primary benchmark for realistic human actions. It contains 13,320 videos across 101 action categories, capturing a wide range of complex temporal dynamics from sports to playing instruments. For this study, we follow the official protocol: models are fine-tuned on the training split and evaluated on the held-out test split. To ensure consistent temporal duration across all evaluation samples, we preprocess the test split into fixed-length 50-frame base clips derived from the original test videos. This step removes variable

video length as a confounding factor, allowing us to isolate the effects of sampling rate and coverage.

Kinetics-400 [8] is utilized to validate scalability and robustness. We fine-tune on the official training split and evaluate on a publicly available labeled test-set release (K4TestSet) hosted on Kaggle [7]. With 400 action classes and approximately 20,000 test videos, it represents a significantly more diverse distribution of human activities. This dataset allows us to test whether aliasing patterns observed in UCF101 hold true in a larger and more unconstrained domain.

3.2 Architectural Paradigms

We selected three state-of-the-art transformer-based architectures, each representing a fundamentally different approach to spatiotemporal modeling. This selection allows us to determine whether aliasing sensitivity arises from intrinsic data properties or architectural inductive biases.

TimeSformer [3] utilizes divided space-time attention, where temporal attention is computed separately from spatial attention. This factorization aggregates global temporal context without requiring dense local continuity. We employ the standard configuration operating on 8-frame inputs.

VideoMAE [13] represents the masked autoencoding paradigm, learning video representations through reconstruction of heavily masked inputs. The architecture follows the commonly adopted 16-frame configuration used in prior work.

ViViT [1] models video as a sequence of spatiotemporal tokens using factorized transformer encoders. We evaluate the standard 32-frame variant balancing global context and token resolution.

The input frame counts ($N = 8, 16, 32$) follow the canonical configurations defined by the original model designs and are constrained by architectural tokenization and positional encoding schemes. Rather than enforcing identical frame counts, which would require modifying pretrained structures and introduce confounding effects, we ensure fairness by controlling the *effective temporal evidence* through the proposed coverage $\times$ stride sampling protocol applied uniformly across all architectures.

3.3 Temporal Sampling Protocol

We developed a systematic sampling protocol to decouple the effects of *temporal coverage* (the span of the action observed) from *temporal stride* (the density of observation). We define a two-dimensional grid of 25 experimental configurations:

- **Coverage Levels:** 10%, 25%, 50%, 75%, and 100% of the total available frames. Reducing coverage simulates scenarios where the sensor is only active for a fraction of the action duration.
- **Stride Values:** 1, 2, 4, 8, and 16 frames between sampled positions. Increasing the stride reduces the sampling rate, directly testing the Nyquist limits of the architecture.

By evaluating every permutation of coverage and stride, we can distinguish between errors caused by missing the action entirely (low coverage) and errors caused by aliasing high-frequency motion (high stride).

3.4 Training and Fine-Tuning Protocol

All architectures were initialized from publicly available Kinetics-400 pretrained checkpoints to ensure a consistent starting representation across models; Kinetics-400 experiments further continue training under our protocol on the official training split.

Fine-tuning was conducted independently for each dataset using the official training splits. Evaluation was performed on the UCF101 test split and on the Kinetics-400 test-set release used in our experiments (K4TestSet, Kaggle).

To ensure fair comparison, identical optimization settings were used across architectures whenever compatible with the original implementations. Models were trained using AdamW with cosine learning-rate decay, an initial learning rate of 5×10^{-5} , batch size 64, and weight decay 0.05. Fine-tuning was conducted for 15 epochs on UCF101 and Kinetics-400.

Spatial augmentations followed standard video-recognition practice, including random resized cropping, horizontal flipping, and normalization using ImageNet statistics. Unless otherwise stated, dense temporal sampling ($\mathcal{C} = 100\%$, $s = 1$) was used during baseline fine-tuning. In additional experiments, we depart from this fixed configuration and apply TRA, where temporal coverage and stride are randomized during training to expose the model to varying sampling conditions. This allows us to study whether robustness to temporal undersampling can be learned rather than only measured during evaluation.

3.5 Evaluation Metrics and Statistical Analysis

To provide a comprehensive assessment, we evaluate models primarily based on Top-1 classification accuracy. Beyond aggregate metrics, we conduct a per-class accuracy analysis to assess heterogeneity across different action types.

Unlike standard benchmarks that report only mean accuracy, we employ a rigorous statistical framework to quantify reliability. We use a two-way Analysis of Variance (ANOVA) to test the main effects of coverage and stride, as well as their interaction, on model performance. Following significant effects, we conduct post-hoc pairwise comparisons using Welch’s t -tests with Bonferroni correction to identify specific thresholds where performance degradation becomes critical. Furthermore, we analyze stability using Levene’s test for equality of variances, assessing whether aggressive downsampling inflates performance variability across action classes.

3.6 Temporal Robustness Augmentation (TRA)

To test whether aliasing-induced brittleness is intrinsic to transformer architectures or instead a consequence of standard dense-sampling training, we introduce **Temporal Robustness Augmentation (TRA)**.

Rather than fine-tuning under a fixed temporal configuration (coverage=100%, stride=1), TRA exposes the model during training to randomized temporal sam-

Table 1: Performance summary under coverage×stride sampling. “Peak” is the best single configuration over the full grid. “Mean@X%” is mean Top-1 accuracy ± std over all strides at coverage X. **Drop** is computed as Mean@100% − Mean@25%.

Dataset	Architecture	Peak Acc.	Mean@100%	Best@100%	Mean@75%	Mean@50%	Mean@25%	Mean@10%	Drop (100→25)
UCF-101	TimeSformer	85.09%	84.22% ± 1.19%	stride-2	84.08%	82.16%	77.36%	73.58%	6.86%
UCF-101	VideoMAE	86.90%	79.85% ± 6.74%	stride-1	74.92%	73.24%	62.68%	54.33%	17.18%
UCF-101	ViViT	85.49%	83.59% ± 2.01%	stride-1	81.49%	78.74%	70.41%	64.05%	13.18%
Kinetics-400	TimeSformer	74.19%	74.01% ± 0.15%	stride-4	73.29%	70.59%	63.41%	56.13%	10.60%
Kinetics-400	VideoMAE	76.52%	75.69% ± 0.26%	stride-1	74.78%	74.59%	68.53%	60.69%	7.16%
Kinetics-400	ViViT	76.19%	76.02% ± 0.27%	stride-1	74.86%	73.10%	67.78%	60.81%	8.23%

pling conditions drawn from the same coverage×stride grid used during evaluation. Specifically, for each training clip, we uniformly sample:

- Coverage $\mathcal{C} \sim \{25\%, 50\%, 75\%, 100\%\}$
- Stride $s \sim \{1, 2, 4, 8, 16\}$

The resulting frame subset is then resized to the model’s required input length. This procedure forces the model to learn temporally global and sampling-invariant representations rather than relying on dense local temporal continuity.

Intuitively, TRA acts as a structured domain randomization in the temporal dimension. By repeatedly exposing the network to aliasing-like conditions during training, we hypothesize that the model learns features that are more stable under undersampling, mitigating catastrophic degradation under sparse observation.

Unless otherwise stated, TRA experiments are conducted using TimeSformer on UCF101 with identical optimization settings as the baseline.

3.7 Computing Infrastructure

All experiments were conducted on a high-performance computing node equipped with dual NVIDIA A100 (40GB) GPUs. Training and evaluation were parallelized using PyTorch Distributed Data Parallel (DDP) to ensure efficiency. To ensure full reproducibility, all evaluation scripts, per-class accuracy logs, and sampling configurations are archived and available in the supplementary material.

4 Results

4.1 Architectural Robustness and Failure Modes

We evaluate Top-1 accuracy across the coverage×stride grid (Table 1). While VideoMAE achieves peak performance on UCF101 (86.9%), it is significantly more fragile than TimeSformer, suffering a 17.18% mean drop at 25% coverage compared to TimeSformer’s 6.86%. Heatmaps (Fig. 4) decouple failure modes into *truncation* (low coverage) and *undersampling* (high stride). VideoMAE displays a sharp gradient along the stride axis, indicating an aliasing-induced failure regime. Conversely, TimeSformer’s factorized attention maintains a stable plateau across moderate strides. Interestingly, TimeSformer exhibits a redundancy effect where stride-2/4 outperforms stride-1 at full coverage, suggesting moderate subsampling acts as a regularizer.

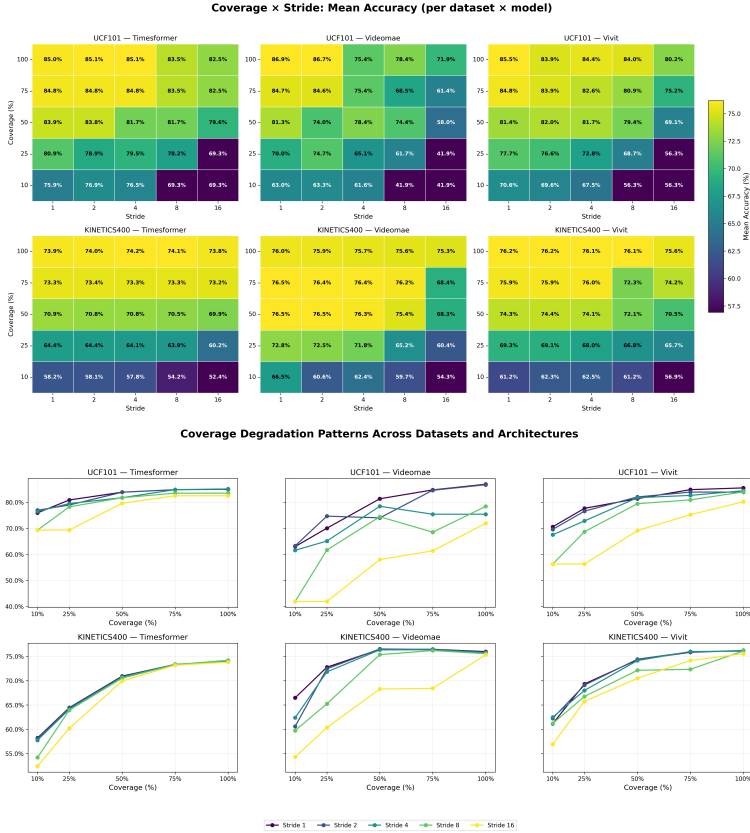


Fig. 4: Temporal Sampling Analysis. Top: Interaction heatmaps showing VideoMAE’s sensitivity to high stride (aliasing) vs. TimeSformer’s stability. Bottom: Accuracy trajectories across architectures; VideoMAE exhibits the steepest collapse as coverage is reduced.

4.2 Statistical Stability and Sensitivity Taxonomy

Accuracy degrades monotonically as temporal evidence is reduced (Fig. 4, bottom), but architectural priors govern the rate. Two-way ANOVA (Table 2) confirms that both factors exert significant main effects ($p < 0.001$). Beyond mean accuracy, Levene’s test reveals significant variance inflation at low coverage ($p < 0.002$), visible in the widening per-class distributions (Fig. 6). This indicates that aggressive downsampling induces instability, with mean accuracy masking catastrophic failure in high-frequency action categories.

To characterize this heterogeneity, we propose an action taxonomy on Kinetics-400 based on aliasing sensitivity (Table 3). We categorize actions into: (i) **High-Sensitivity** ($\Delta > 20\%$): explosive, short events (e.g., jumping jacks, diving) where missing critical frames is destructive; (ii) **Moderate (10–20%)**: dynamic motion with partial redundancy; and (iii) **Low-Sensitivity** ($\Delta \leq 10\%$): low-frequency or context-dominated actions (e.g., typing, billiards). Figure 5 confirms

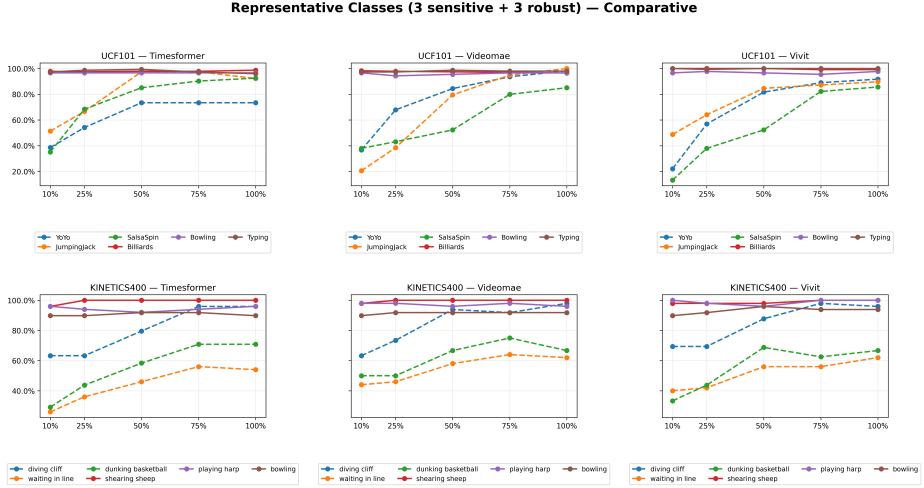


Fig. 5: Accuracy trajectories for representative action classes selected by aliasing sensitivity.

these tiers, showing that sensitive actions suffer both steeper degradation and higher dispersion than robust ones.

Table 2: ANOVA Effect Sizes (η^2) and Levene Stability Test.

Dataset	Model	Cov. η^2	Stride η^2	Levene p	d_Δ
UCF-101	TSF	0.061	0.010	0.002	0.63
	VMAE	0.206	0.087	<0.001	1.38
	ViViT	0.143	0.030	<0.001	1.05
Kinetics-400	TSF	0.136	0.002	0.011	1.04
	VMAE	0.117	0.023	0.001	0.83
	ViViT	0.072	0.004	0.029	0.78

Table 3: Kinetics-400 action taxonomy by aliasing sensitivity (Δ), computed as Mean@100% – Mean@25% (averaged over strides).

Tier	Δ	Count	Typical Characteristics
High-Sens.	> 20%	107	Explosive, high-velocity, short events
Moderate	10–20%	193	Dynamic, partially redundant motion
Low-Sens.	$\leq$ 10%	100	Low-frequency, redundant, or context-dominated

4.3 Spectral Validation of Nyquist–Shannon Sampling

To move beyond empirical observation, we provide a quantitative validation of the Nyquist-Shannon sampling principle in video action recognition by modeling action dynamics as temporal signals derived from optical-flow magnitude. Each action class can therefore be characterized by its dominant motion frequency.

High-sensitivity actions such as *YoYo* and *JumpingJack* exhibit dominant frequencies (f_{peak}) between 5.8 Hz to 6.5 Hz, whereas temporally redundant activities including *Billiards* and *Bowling* concentrate most spectral energy in low-

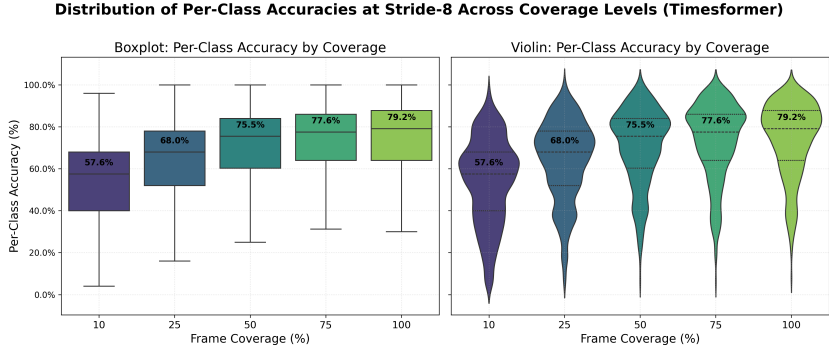


Fig. 6: Per-class distributions at stride=8. Lower coverage (light) inflates dispersion.

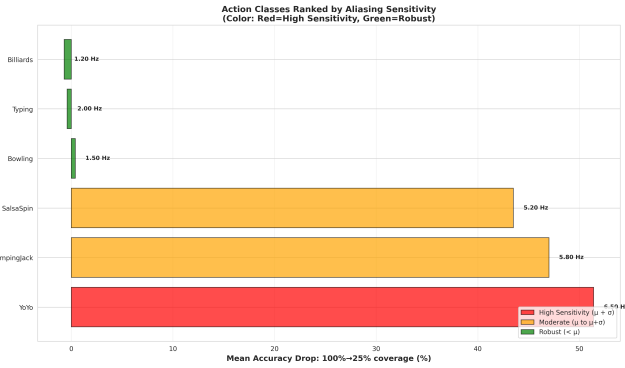


Fig. 7: Action classes ranked by aliasing sensitivity. Labels indicate the dominant frequency in Hz, illustrating that high-frequency motion (e.g., YoYo at 6.50 Hz) is more susceptible to performance drops than low-frequency actions (e.g., Billiards at 1.20 Hz).

frequency bands (1.2 Hz to 1.5 Hz). This distinction reveals that actions differ substantially in their intrinsic temporal bandwidth.

Across action classes, dominant motion frequency exhibits a strong monotonic relationship with recognition degradation under temporal subsampling (Pearson $r = 0.99$, Spearman $\rho = 0.94$, $p < 0.01$), while low-frequency energy shows an inverse correlation with performance drop. These results indicate that architectural brittleness emerges when the effective sampling rate f_s falls below approximately twice the dominant motion frequency, consistent with Nyquist sampling limits.

The relationship between spectral content and recognition robustness is further quantified in Table 4. We observe a near-perfect Spearman correlation ($\rho = 0.943$, $p < 0.01$) between the dominant frequency and the mean accuracy drop. This provides strong evidence that the brittleness observed in architectures such as VideoMAE is consistent with Nyquist-style undersampling effects: when the effective sampling rate f_s approaches or falls below approximately twice the dominant motion frequency, discriminative motion cues become unreliable due to aliasing.

Table 4: Correlation between Spectral Metrics and Accuracy Drop

Metric	Pearson r	Spearman ρ	Interpretation
Dominant Freq.	0.991	0.943	Strong Positive
Spectral Centroid	0.995	0.995	Strong Positive
Low-Freq Energy	-0.965	-0.964	Strong Negative
Spectral Flatness	0.962	0.961	Strong Positive

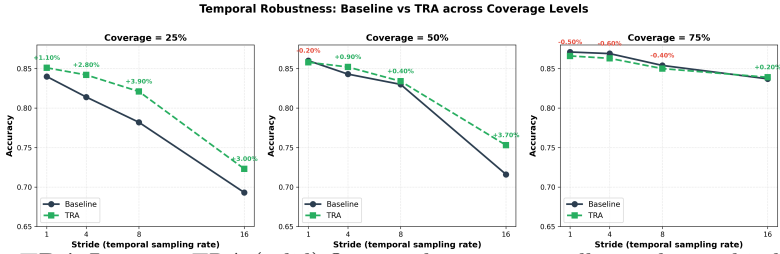


Fig. 8: TRA Impact. TRA (solid) flattens the accuracy collapse observed in baseline (dashed) at low coverage and high stride.

These findings, summarized in Fig. 7, categorize actions into a clear hierarchy: high-frequency actions are "aliasing-prone," while low-frequency actions are "aliasing-robust". This supports our conclusion that temporal sampling should be content-adaptive, tailoring the resolution to the intrinsic frequency of the observed activity.

4.4 Mitigating Aliasing via Temporal Robustness Augmentation

To determine whether temporal aliasing is a fundamental sensing limitation or partly a train-time artifact, we evaluate TimeSformer with Temporal Robustness Augmentation (TRA). As shown in Fig. 8, TRA consistently reduces the degradation observed under sparse temporal sampling.

The most pronounced improvements occur at lower coverage levels and higher strides. For example, at 25% coverage with stride $s = 8$, accuracy improves from 0.782 to 0.821 (+5.0%), while at stride $s = 16$ performance increases from 0.693 to 0.723 (+4.3%). Moderate improvements are also observed at 50% coverage, including a +5.2% gain at stride $s = 16$.

Importantly, these robustness gains do not come at the cost of dense-sampling performance. At full temporal coverage (100%), TRA maintains nearly identical accuracy to the baseline across all strides, with differences below 0.7%. This indicates that exposure to variable temporal sampling during training encourages the model to learn representations that are less dependent on dense frame continuity while preserving performance under standard conditions.

Interpretation. TRA does not increase the physical sampling rate but alters the representation to prioritize lower-frequency, semantically stable motion components. This effectively yields an aliasing-aware model that remains robust under aggressive undersampling, mitigating signal distortion through representation learning rather than increased sensing.

Table 5: TimeSformer Baseline vs. TRA on UCF101.

Model	S	Cov. 25%		Cov. 50%		Cov. 100%	
		Base	TRA ($\Delta\%$)	Base	TRA ($\Delta\%$)	Base	TRA ($\Delta\%$)
TSF	1	0.840	0.851 (+1.3)	0.860	0.858 (-0.3)	0.871	0.866 (-0.7)
	4	0.814	0.842 (+3.5)	0.843	0.852 (+1.1)	0.869	0.863 (-0.6)
	8	0.782	0.821 (+5.0)	0.830	0.834 (+0.5)	0.854	0.850 (-0.4)
	16	0.693	0.723 (+4.3)	0.716	0.753 (+5.2)	0.837	0.839 (+0.2)

5 Discussion

5.1 Architectural Resilience vs. Brittleness

Our results expose a fundamental trade-off between *peak capacity* under ideal sampling and *robustness* under constrained observation.

Brittleness of reconstruction-based modeling. VideoMAE consistently achieves the highest peak accuracy but suffers the most severe degradation. Its 17.18% drop on UCF101 stems from a masked autoencoding objective that relies on dense local temporal correlations to reconstruct patches. When stride increases or coverage decreases, these correlations are violated, leaving the model with insufficient information to infer motion semantics. Consequently, reconstruction-based representations appear structurally fragile under undersampling.

Robustness induced by factorized attention. Conversely, TimeSformer exhibits significantly greater resilience. Its divided space-time attention aggregates evidence globally rather than relying on frame-to-frame continuity. This factorization acts as an implicit low-pass filter, suppressing high-frequency artifacts and preserving semantic cues even near critical sampling limits. ViViT occupies an intermediate regime, suggesting that inductive biases in tokenization and attention factorization are primary drivers of aliasing behavior.

5.2 Empirical Validation of Nyquist–Shannon

Our sensitivity taxonomy (Table 3) provides empirical evidence that action recognition obeys classical signal-processing principles. High-sensitivity actions (e.g., diving) involve short-lived, high-frequency events that collapse when discriminative cues fall between sparse sampling intervals. In contrast, low-sensitivity actions (e.g., typing) possess enough temporal redundancy to remain identifiable below the effective Nyquist limit. The variance explosion observed at low coverage (Figure 6) further indicates that recognition becomes stochastic when the sampling grid fails to align with critical frames, a failure mode often masked by mean accuracy.

5.3 Implications for System Design

These findings offer three key insights for deployment:

- 1. Architecture Selection:** In power-limited environments (e.g., robotics), global temporal aggregation models like TimeSformer provide a safer operating margin than reconstruction-based models.
- 2. Adaptive Sampling:** Fixed-rate sampling is inefficient; systems should dynamically reallocate resolution where high-frequency motion is detected.

3. Robustness Training: Standard pipelines assume dense availability. Incorporating variable-stride augmentation (TRA) encourages models to learn globally stable representations, mitigating brittleness without architectural changes.

5.4 Is Aliasing Fundamental or Learnable?

The observed ($\sim 5\%$) improvements from TRA suggest that temporal brittleness is, in part, a train-time bias. While sampling theory imposes limits on signal reconstruction, classification objectives permit partial compensation through invariant feature learning. Temporal sampling should thus be viewed as both a sensing constraint and a learnable representation problem.

5.5 Limitations and Future Directions

While we estimate frequency descriptors from optical-flow magnitude, our spectral analysis is limited to a specific motion proxy and preprocessing pipeline. Future work should evaluate alternative frequency estimators (e.g., standardized PSD protocols, pose trajectories, or class-conditioned flow fields) and establish more formal sampling guarantees. Additionally, exploring hierarchical architectures (e.g., MViT) could determine whether multi-scale representations better reconcile peak performance with robustness. Finally, our experiments simulate temporal undersampling from videos originally captured at higher frame rates; studying the effects of low-frequency data collection at the sensor level remains an important direction for future work.

6 Conclusion

This work establishes temporal sampling as a first-class research variable in action recognition. Through a systematic coverage-by-stride stress test across two large-scale benchmarks, we provide empirical evidence that modern architectures exhibit predictable responses to temporal undersampling governed by classical signal processing principles. Our results uncover a fundamental trade-off: reconstruction-based models like VideoMAE achieve superior peak performance but are structurally brittle under aliasing, whereas factorized attention models like TimeSformer maintain stability across diverse sampling regimes.

Beyond mean accuracy, we show that aggressive temporal reduction induces significant variance inflation, revealing instability in high-frequency action classes that is often obscured by aggregate metrics. By introducing an action sensitivity taxonomy, we provide a principled framework for reasoning about spatiotemporal resolution in sensing systems. Finally, we demonstrate that aliasing-induced degradation can be partially mitigated through TRA, which improves robustness under sparse sampling regimes (up to $\sim 5\%$ relative improvement in our experiments) by encouraging sampling-invariant representations. These findings motivate a shift toward content-adaptive, signal-processing-guided video understanding that prioritizes resolution where it is information-critical.

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